

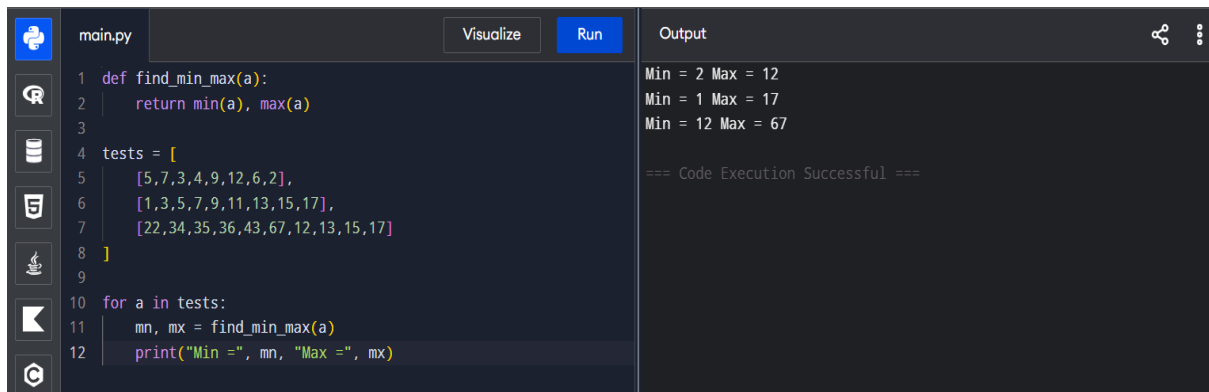
CHAPTER – 3

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EXPERIMENT – 1



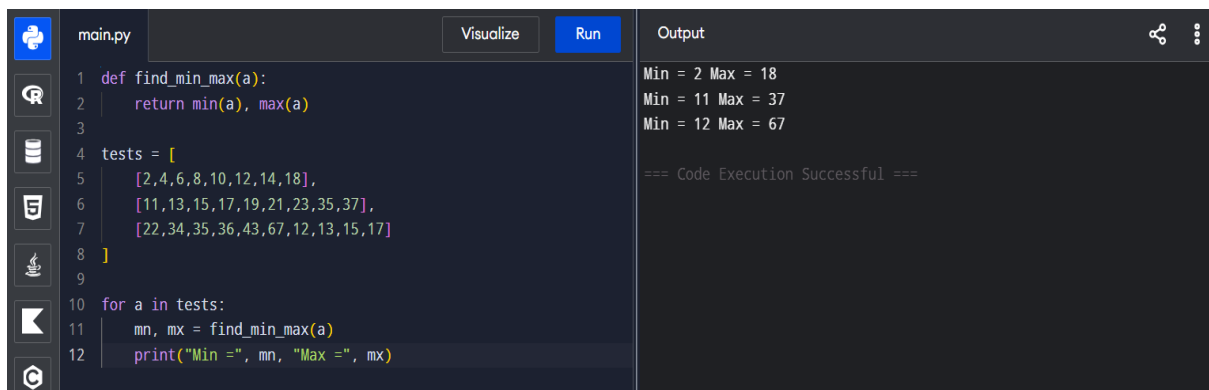
The screenshot shows a Python IDE with a file named 'main.py'. The code defines a function 'find_min_max(a)' that returns the minimum and maximum values of a list 'a'. It then tests this function with three different lists. The output shows the results for each list.

```
1 def find_min_max(a):
2     return min(a), max(a)
3
4 tests = [
5     [5,7,3,4,9,12,6,2],
6     [1,3,5,7,9,11,13,15,17],
7     [22,34,35,36,43,67,12,13,15,17]
8 ]
9
10 for a in tests:
11     mn, mx = find_min_max(a)
12     print("Min =", mn, "Max =", mx)
```

Output:

```
Min = 2 Max = 12
Min = 1 Max = 17
Min = 12 Max = 67
=== Code Execution Successful ===
```

EXPERIMENT – 2



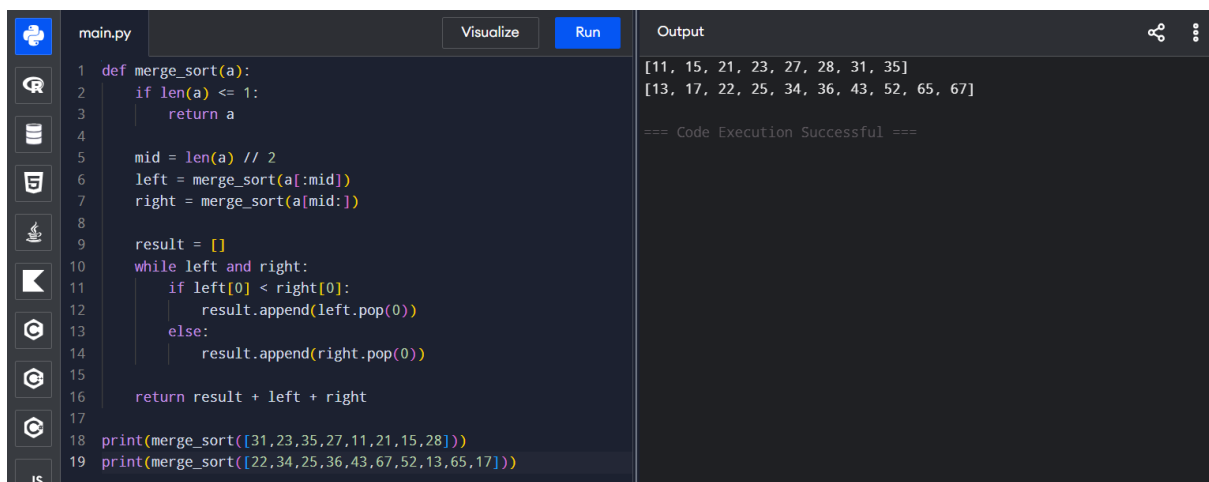
The screenshot shows a Python IDE with a file named 'main.py'. The code defines a function 'find_min_max(a)' that returns the minimum and maximum values of a list 'a'. It then tests this function with three different lists. The output shows the results for each list.

```
1 def find_min_max(a):
2     return min(a), max(a)
3
4 tests = [
5     [2,4,6,8,10,12,14,18],
6     [11,13,15,17,19,21,23,35,37],
7     [22,34,35,36,43,67,12,13,15,17]
8 ]
9
10 for a in tests:
11     mn, mx = find_min_max(a)
12     print("Min =", mn, "Max =", mx)
```

Output:

```
Min = 2 Max = 18
Min = 11 Max = 37
Min = 12 Max = 67
=== Code Execution Successful ===
```

EXPERIMENT – 3



The screenshot shows a Python IDE with a file named 'main.py'. The code defines a 'merge_sort(a)' function using a divide-and-conquer approach. It then tests the function with two different lists. The output shows the sorted lists.

```
1 def merge_sort(a):
2     if len(a) <= 1:
3         return a
4
5     mid = len(a) // 2
6     left = merge_sort(a[:mid])
7     right = merge_sort(a[mid:])
8
9     result = []
10    while left and right:
11        if left[0] < right[0]:
12            result.append(left.pop(0))
13        else:
14            result.append(right.pop(0))
15
16    return result + left + right
17
18 print(merge_sort([31,23,35,27,11,21,15,28]))
19 print(merge_sort([22,34,25,36,43,67,52,13,65,17]))
```

Output:

```
[11, 15, 21, 23, 27, 28, 31, 35]
[13, 17, 22, 25, 34, 36, 43, 52, 65, 67]
=== Code Execution Successful ===
```

EXPERIMENT – 4

main.py

Visualize

Run

1

def merge_sort(a):

2

global count

3

if len(a) <= 1:

4

return a

5

m = len(a)//2

6

left = merge_sort(a[:m])

7

right = merge_sort(a[m:])

8

result = []

9

while left and right:

10

count += 1

11

if left[0] <= right[0]:

12

result.append(left.pop(0))

13

else:

14

result.append(right.pop(0))

15

return result + left + right

16

tests = []

17

[12,4,78,23,45,67,89,1],

18

[38,27,43,3,9,82,10]

19

]

20

for a in tests:

21

count = 0

22

print("Sorted:", merge_sort(a))

Output

Sorted: [1, 4, 12, 23, 45, 67, 78, 89]

Comparisons: 16

Sorted: [3, 9, 10, 27, 38, 43, 82]

Comparisons: 13

=== Code Execution Successful ===

EXPERIMENT – 5

main.py

Visualize

Run

1

def quick_sort(a):

2

if len(a) <= 1:

3

return a

4

pivot = a[0]

5

left = [x for x in a[1:] if x <= pivot]

6

right = [x for x in a[1:] if x > pivot]

7

print("Pivot:", pivot, "->", left + [pivot] + right)

8

return quick_sort(left) + [pivot] + quick_sort(right)

9

tests = [

10

[10,16,8,12,15,6,3,9,5],

11

[12,4,78,23,45,67,89,1],

12

[38,27,43,3,9,82,10]

13

]

14

for a in tests:

15

print("Sorted:", quick_sort(a))

16

print()

Output

Pivot: 10 -> [8, 6, 3, 9, 5, 10, 16, 12, 15]

Pivot: 8 -> [6, 3, 5, 8, 9]

Pivot: 6 -> [3, 5, 6]

Pivot: 3 -> [3, 5]

Pivot: 16 -> [12, 15, 16]

Pivot: 12 -> [12, 15]

Sorted: [3, 5, 6, 8, 9, 10, 12, 15, 16]

Pivot: 12 -> [4, 1, 12, 78, 23, 45, 67, 89]

Pivot: 4 -> [1, 4]

Pivot: 78 -> [23, 45, 67, 78, 89]

Pivot: 23 -> [23, 45, 67]

Pivot: 45 -> [45, 67]

Sorted: [1, 4, 12, 23, 45, 67, 78, 89]

Pivot: 38 -> [27, 3, 9, 10, 38, 43, 82]

Pivot: 27 -> [3, 9, 10, 27]

Pivot: 3 -> [3, 9, 10]

Pivot: 9 -> [9, 10]

Pivot: 43 -> [43, 82]

Sorted: [3, 9, 10, 27, 38, 43, 82]

EXPERIMENT – 6

main.py

Visualize

Run

1

def quick_sort(a):

2

if len(a) <= 1:

3

return a

4

pivot = a[len(a)//2]

5

left = [x for x in a if x < pivot]

6

mid = [x for x in a if x == pivot]

7

right = [x for x in a if x > pivot]

8

print("Pivot:", pivot, "->", left + mid + right)

9

return quick_sort(left) + mid + quick_sort(right)

10

tests = [

11

[19,72,35,46,58,91,22,31],

12

[31,23,35,27,11,21,15,28],

13

[22,34,25,36,43,67,52,13,65,17]

14

]

15

for a in tests:

16

print("Sorted:", quick_sort(a))

17

print()

Output

Pivot: 58 -> [19, 35, 46, 22, 31, 58, 72, 91]

Pivot: 46 -> [19, 35, 22, 31, 46]

Pivot: 22 -> [19, 22, 35, 31]

Pivot: 31 -> [31, 35]

Pivot: 91 -> [72, 91]

Sorted: [19, 22, 31, 35, 46, 58, 72, 91]

Pivot: 11 -> [11, 31, 23, 35, 27, 21, 15, 28]

Pivot: 27 -> [23, 21, 15, 27, 31, 35, 28]

Pivot: 21 -> [15, 21, 23]

Pivot: 35 -> [31, 28, 35]

Pivot: 28 -> [28, 31]

Sorted: [11, 15, 21, 23, 27, 28, 31, 35]

Pivot: 67 -> [22, 34, 25, 36, 43, 52, 13, 65, 17, 67]

Pivot: 43 -> [22, 34, 25, 36, 13, 17, 43, 52, 65]

Pivot: 36 -> [22, 34, 25, 13, 17, 36]

Pivot: 25 -> [22, 13, 17, 25, 34]

Pivot: 13 -> [13, 22, 17]

Pivot: 17 -> [17, 22]

Pivot: 65 -> [52, 65]

Sorted: [13, 17, 22, 25, 34, 36, 43, 52, 65, 67]

EXPERIMENT – 7

main.py	Visualize	Run	Output
<pre>1 def binary_search(a, key): 2 low, high = 0, len(a) - 1 3 count = 0 4 while low <= high: 5 mid = (low + high) // 2 6 count += 1 7 if a[mid] == key: 8 return mid + 1, count 9 elif key < a[mid]: 10 high = mid - 1 11 else: 12 low = mid + 1 13 return -1, count 14 tests = [15 ([5,10,15,20,25,30,35,40,45], 20), 16 ([10,20,30,40,50,60], 50), 17 ([21,32,40,54,65,76,87], 32) 18] 19 for a, key in tests: 20 pos, comparisons = binary_search(a, key) 21 print("Position:", pos) 22 print("Comparisons:", comparisons)</pre>			<pre>Position: 4 Comparisons: 4 Position: 5 Comparisons: 2 Position: 2 Comparisons: 2 === Code Execution Successful ===</pre>

EXPERIMENT – 8

main.py	Visualize	Run	Output
<pre>1 def binary_search(a, key): 2 low, high = 0, len(a) - 1 3 while low <= high: 4 mid = (low + high) // 2 5 print("Low:", low, "High:", high, "Mid:", mid) 6 if a[mid] == key: 7 return mid + 1 8 elif key < a[mid]: 9 high = mid - 1 10 else: 11 low = mid + 1 12 return -1 13 tests = [14 ([3,9,14,19,25,31,42,47,53], 31), 15 ([13,19,24,29,35,41,42], 42), 16 ([20,40,60,80,100,120], 60) 17] 18 for a, key in tests: 19 print("\nArray:", a) 20 print("Position:", binary_search(a, key))</pre>			<pre>Array: [3, 9, 14, 19, 25, 31, 42, 47, 53] Low: 0 High: 8 Mid: 4 Low: 5 High: 8 Mid: 6 Low: 5 High: 5 Mid: 5 Position: 6 Array: [13, 19, 24, 29, 35, 41, 42] Low: 0 High: 6 Mid: 3 Low: 4 High: 6 Mid: 5 Low: 6 High: 6 Mid: 6 Position: 7 Array: [20, 40, 60, 80, 100, 120] Low: 0 High: 5 Mid: 2 Position: 3 === Code Execution Successful ===</pre>

EXPERIMENT – 9

main.py	Visualize	Run	Output
<pre>1 def k_closest(points, k): 2 points.sort(key=lambda p: p[0]**2 + p[1]**2) 3 return points[:k] 4 5 tests = [6 ([[1,3],[-2,2],[5,8],[0,1]], 2), 7 ([[1,3],[-2,2]], 1), 8 ([[3,3],[5,-1],[-2,4]], 2) 9] 10 11 for p, k in tests: 12 print(k_closest(p, k))</pre>			<pre>[[0, 1], [-2, 2]] [[-2, 2]] [[3, 3], [-2, 4]] === Code Execution Successful ===</pre>

EXPERIMENT – 10

```
main.py Visualize Run Output
1 def four_sum(A, B, C, D):
2     count = 0
3
4     for a in A:
5         for b in B:
6             for c in C:
7                 for d in D:
8                     if a + b + c + d == 0:
9                         count += 1
10
11     return count
12
13 print(four_sum([1,2], [-2,-1], [-1,2], [0,2]))
14 print(four_sum([0], [0], [0], [0]))
```

```
2
1

=== Code Execution Successful ===
```

EXPERIMENT – 11

```
main.py Visualize Run Output
1 def median_medians(a, k):
2     if len(a) <= 5:
3         return sorted(a)[k - 1]
4     groups = [a[i:i+5] for i in range(0, len(a), 5)]
5     medians = [sorted(g)[len(g)//2] for g in groups]
6     pivot = median_medians(medians, (len(medians)+1)//2)
7     low = [x for x in a if x < pivot]
8     high = [x for x in a if x > pivot]
9     equal = [x for x in a if x == pivot]
10    if k <= len(low):
11        return median_medians(low, k)
12    elif k <= len(low) + len(equal):
13        return pivot
14    else:
15        return median_medians(high, k-len(low)-len(equal))
16 print(median_medians([12,3,5,7,19], 2))
17 print(median_medians([12,3,5,7,4,19,26], 3))
18 print(median_medians([1,2,3,4,5,6,7,8,9,10], 6))
```

```
5
5
6

=== Code Execution Successful ===
```

EXPERIMENT – 12

```
main.py Visualize Run Output
1 def median_of_medians(a, k):
2     if len(a) <= 5:
3         return sorted(a)[k-1]
4     groups = [a[i:i+5] for i in range(0, len(a), 5)]
5     med = [sorted(g)[len(g)//2] for g in groups]
6     pivot = sorted(med)[len(med)//2]
7
8     low = [x for x in a if x < pivot]
9     high = [x for x in a if x > pivot]
10
11    if k <= len(low):
12        return median_of_medians(low, k)
13    elif k == len(low) + 1:
14        return pivot
15    else:
16        return median_of_medians(high, k-len(low)-1)
17
18 print(median_of_medians([1,2,3,4,5,6,7,8,9,10], 6))
19 print(median_of_medians([23,17,31,44,55,21,20,18,19,27], 5))
```

```
6
21

=== Code Execution Successful ===
```

EXPERIMENT – 13

```
main.py Visualize Run Output
1 from itertools import combinations
2 def meet_middle(a, target):
3     n = len(a)
4     mid = n // 2
5     left = a[:mid]
6     right = a[mid:]
7     L = []
8     R = []
9     for r in range(len(left)+1):
10         for c in combinations(left, r):
11             L.append((sum(c), c))
12     for r in range(len(right)+1):
13         for c in combinations(right, r):
14             R.append((sum(c), c))
15     best = (float('inf'), ())
16     for s1, c1 in L:
17         for s2, c2 in R:
18             diff = abs(target - (s1+s2))
19             if diff < best[0]:
20                 best = (diff, c1+c2)
21     return best[1], sum(best[1])
22 print(meet_middle([45,34,4,12,5,2], 42))
```

Output

```
((34, 5, 2), 41)
((4, 6), 10)

=== Code Execution Successful ===
```

EXPERIMENT – 14

```
main.py Visualize Run Output
1 def meet_middle(a, target):
2     n = len(a)
3     mid = n // 2
4     left = a[:mid]
5     right = a[mid:]
6     sums = set()
7     for mask in range(1 << len(left)):
8         s = sum(left[i] for i in range(len(left)) if mask &
9             (1 << i))
10        sums.add(s)
11    for mask in range(1 << len(right)):
12        s = sum(right[i] for i in range(len(right)) if mask &
13            (1 << i))
14        if target - s in sums:
15            return True
16    return False
17 print(meet_middle([1,3,9,2,7,12], 15))
18 print(meet_middle([3,34,4,12,5,2], 15))
```

Output

```
True
True

=== Code Execution Successful ===
```

EXPERIMENT – 15

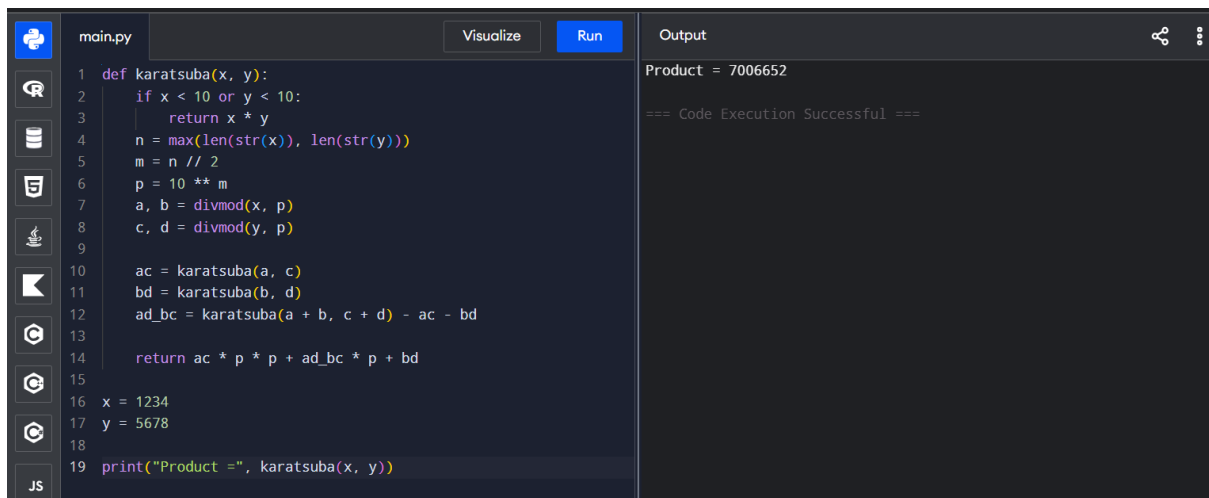
```
main.py Visualize Run Output
1 def strassen(A, B):
2     a, b = A[0]
3     c, d = A[1]
4     e, f = B[0]
5     g, h = B[1]
6     p1 = a * (f - h)
7     p2 = (a + b) * h
8     p3 = (c + d) * e
9     p4 = d * (g - e)
10    p5 = (a + d) * (e + h)
11    p6 = (b - d) * (g + h)
12    p7 = (a - c) * (e + f)
13    return [
14        [p5 + p4 - p2 + p6, p1 + p2],
15        [p3 + p4, p1 + p5 - p3 - p7]
16    ]
17 A = [[1, 7], [3, 5]]
18 B = [[6, 8], [4, 2]]
19
20 print(strassen(A, B))
```

Output

```
[[34, 22], [38, 34]]

=== Code Execution Successful ===
```

EXPERIMENT – 16



```
1 def karatsuba(x, y):
2     if x < 10 or y < 10:
3         return x * y
4     n = max(len(str(x)), len(str(y)))
5     m = n // 2
6     p = 10 ** m
7     a, b = divmod(x, p)
8     c, d = divmod(y, p)
9
10    ac = karatsuba(a, c)
11    bd = karatsuba(b, d)
12    ad_bc = karatsuba(a + b, c + d) - ac - bd
13
14    return ac * p * p + ad_bc * p + bd
15
16 x = 1234
17 y = 5678
18
19 print("Product =", karatsuba(x, y))
```

Output

Product = 7006652

=== Code Execution Successful ===